

Time Series Analysis

Seminar 3: Unit roots and ARIMA models



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Acronyms used in this material

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **DF**: Dickey–Fuller (test)
- **DW**: Durbin–Watson (statistic)
- **EODHD**: EOD Historical Data (market-data provider)
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices
- **KPSS**: Kwiatkowski–Phillips–Schmidt–Shin (test)
- **MA**: Moving Average
- **OLS**: Ordinary Least Squares
- **PP**: Phillips–Perron (test)
- **SE**: Standard Error
- **SES**: Simple Exponential Smoothing
- **US**: United States
- **ZA**: Zivot–Andrews (test)

Today's Question and Seminar Structure

- The question of the seminar
  - **does a series have a unit root, how many times must we difference it, and how wide are the forecast intervals of the resulting ARIMA model?**
- Results to obtain (nothing is handed in)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C2
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	a Dickey–Fuller statistic by hand; choosing the deterministic terms	Solved, Proposed	A1
A3, A4	forecasts and intervals of ARIMA(1,1,0) and ARIMA(0,1,1)	Solved, Proposed	A3
A5, A6	the order of integration read from an equation	Solved, Proposed	A5
B1, B2	unit-root tests: BET; S&P 500 and EUR/RON	Solved, Proposed	B1
B3, B4	ARIMA: Romanian real GDP; Romanian inflation	Solved, Proposed	B3
C1, C2	a break or a unit root? what is wrong in an AI answer?	Proposed	B1, B3

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in class
- Part A: calculations on paper; Part B: real data in the notebook, each task ending with an interpretation question; Part C: open questions and the critique of an AI answer

Data used

Series	Source	Frequency	Period
BET, S&P 500	EODHD	daily, close	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026
Real GDP, Romania	Eurostat (namq_10_gdp)	quarterly, seasonally adjusted	2000–2026
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	2004–2026

- Prices and GDP as $100 \ln$: differences are then log returns or growth rates in %; 12-month inflation $\pi_t = 100(\ln P_t - \ln P_{t-12})$
- In the notebook: `adf_test`, `pp_test`, `kpss_test`, `za_test`, `arima_grid`; no account or key is needed

Prerequisites for Today (1/16): two kinds of trend

- **Trend-stationary**: a deterministic line plus stationary deviations

$$y_t = \alpha + \beta t + u_t, \qquad u_t \text{ stationary}$$

  - α: the intercept; β: the slope, the average change per period; u_t: the deviation from the line
  - shocks fade: the series returns to the line

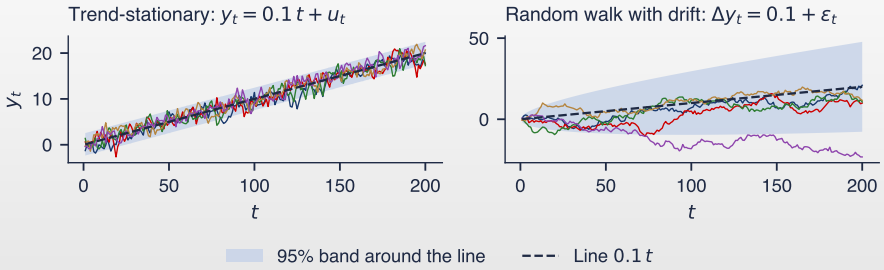
- **Difference-stationary**: the changes are stationary

$$\Delta y_t = y_t - y_{t-1} = \beta + u_t$$

  - u_t white noise: the random walk with drift β
  - shocks are permanent: each one shifts all later values; there is no line to return to

- Both have the same average path $\alpha + \beta t$ (with $y_0 = \alpha$); they differ in how the uncertainty grows

Prerequisites for Today (2/16): the two trends on simulated paths



- Five simulated paths of each model, the same line $0.1\,t$, $T = 200$
- The 95% band around the line
  - trend-stationary (u_t an AR(1), $\phi = 0.6$): constant width, ± 2.45; every path stays near the line
  - random walk with drift: the band widens like $\pm 1.96\sqrt{t}$, ± 27.7 at $t = 200$; the paths drift apart

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Prerequisites for Today (3/16): unit roots and the order of integration

- **Unit root**: the AR polynomial $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$ has the root $z = 1$

$$\phi(z) = (1 - z)\,\phi^*(z) \quad \Longrightarrow \quad \phi^*(L)\,\Delta y_t = \varepsilon_t$$

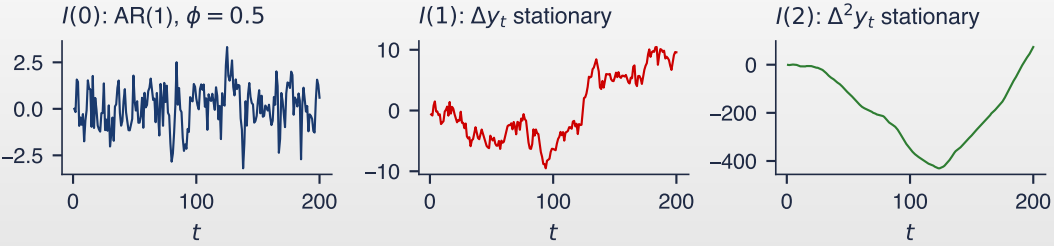
  - $\phi^*(z)$: the remaining polynomial, of degree $p - 1$; L: the lag operator, $L y_t = y_{t-1}$
  - the factor $1 - L$ is the difference: the differences follow a stationary AR($p - 1$) if all roots of ϕ^* have $|z| > 1$
  - a stationary AR has all roots with $|z| > 1$ (Chapter 2)

- **Integrated of order** d, $y_t \sim I(d)$: $\Delta^d y_t$ is stationary, $\Delta^{d-1} y_t$ is not

$$\Delta y_t = y_t - y_{t-1}, \qquad \Delta^2 y_t = \Delta y_t - \Delta y_{t-1} = y_t - 2y_{t-1} + y_{t-2}$$

  - d: the number of differences needed to reach stationarity; $I(0)$: stationary
  - typical: log prices and log GDP are $I(1)$; returns and growth rates are $I(0)$

Prerequisites for Today (4/16): three orders of integration



- The same simulated shocks, $T = 200$, summed zero, one and two times
- How each one looks
  - $I(0)$: oscillates around a fixed mean
  - $I(1)$: wanders, with no mean to return to; its changes look like the left panel
  - $I(2)$: smooth, with long swings in its slope; two differences are needed

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Prerequisites for Today (5/16): spurious regression

- **Regression** of one series on another, by OLS (ordinary least squares)

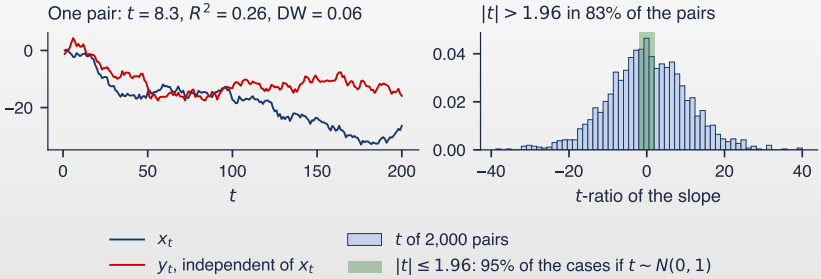
$$y_t = a + b\,x_t + e_t, \qquad t = \frac{\hat{b}}{\mathrm{SE}(\hat{b})}, \qquad R^2 = 1 - \frac{\sum_t \hat{e}_t^2}{\sum_t (y_t - \bar{y})^2}$$

  - t: the t-ratio of the slope, compared with ± 1.96; R^2: the share of the variance of y_t explained by x_t
- **Spurious regression** Granger and Newbold (1974): two independent $I(1)$ series often give a large $|t|$ and R^2
  - the residuals are themselves $I(1)$, so the usual t distribution does not hold
- **Durbin–Watson** (DW): a check of the residual autocorrelation

$$DW = \frac{\sum_{t=2}^{T} (\hat{e}_t - \hat{e}_{t-1})^2}{\sum_{t=1}^{T} \hat{e}_t^2} \approx 2\,(1 - \hat{\rho}_{\hat{e}}(1))$$

  - $DW \approx 2$: uncorrelated residuals; DW near 0: very persistent residuals
  - warning sign of a spurious regression: $R^2 > DW$; the fix: regress the differences

Prerequisites for Today (6/16): a spurious regression, simulated



- Left: two independent random walks, $T = 200$; the regression of y_t on x_t gives $t = 8.3$, $R^2 = 0.26$, $DW = 0.06$
- Right: 2 000 such pairs; the t-ratio lies outside ± 1.96 in 83% of them, instead of 5%
- $R^2 = 0.26 > DW = 0.06$: the warning sign of the previous slide

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Prerequisites for Today (7/16): the Dickey–Fuller test

- **ADF regression** (augmented Dickey–Fuller) Dickey and Fuller (1979)

$$\Delta y_t = c + bt + \gamma y_{t-\mathbf{1}} + \sum_{j=\mathbf{1}}^{k} \delta_j \Delta y_{t-j} + \varepsilon_t, \qquad \gamma = \phi - 1$$

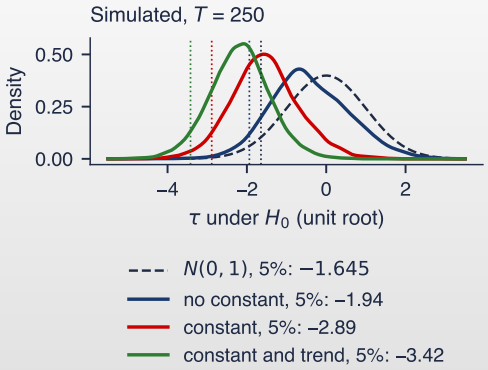
  - c: constant; bt: linear trend; δ_j: coefficients of the past differences Δy_{t-j}, which absorb the short-run dynamics; k: their number
  - ϕ: the AR(1) coefficient of y_t (with $k = 0$, $y_t = c + bt + \phi y_{t-\mathbf{1}} + \varepsilon_t$)

- **Hypotheses** and **statistic**

$$H_{\mathbf{0}}\colon \gamma = 0 \text{ (unit root)}, \qquad H_{\mathbf{1}}\colon \gamma < 0 \text{ (stationary)}, \qquad \tau = \frac{\hat{\gamma}}{\mathrm{SE}(\hat{\gamma})}$$

  - SE: the standard error; reject $H_{\mathbf{0}}$ if τ is below (more negative than) the critical value
  - $k = 0$: the Dickey–Fuller (DF) test; k chosen by AIC, at most $12(T/100)^{\mathbf{1/4}}$

Prerequisites for Today (8/16): the Dickey–Fuller distributions



- **Under H_0, τ is not Student t**: its distribution is shifted to the left
  - chart: 20 000 simulated random walks, $T = 250$; 5% quantiles -1.94, -2.89, -3.42
  - the more deterministic terms, the further to the left
- **5% critical values** for large T MacKinnon (1994)
  - no constant -1.94; constant -2.86; constant and trend -3.41
  - the Normal value -1.645 would reject the unit root far too often
- **Deterministic terms**: chosen from the plot
  - trend for trending series (log GDP, log prices); constant for series around a level (inflation, interest rates, returns)

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Prerequisites for Today (9/16): long-run variance and the Phillips–Perron test

- **Long-run variance** of a stationary series u_t: the variance plus all autocovariances

$$\omega^2 = \gamma(0) + 2\sum_{h=1}^{\infty}\gamma(h), \qquad \hat{\omega}^2 = \hat{\gamma}(0) + 2\sum_{h=1}^{\ell}\Big(1 - \frac{h}{\ell+1}\Big)\hat{\gamma}(h)$$

  - ω^2/T is the variance of the mean $\bar{u}$ when the u_t are autocorrelated; for white noise $\omega^2 = \gamma(0)$
  - $\hat{\omega}^2$: the Newey–West estimator; ℓ: the number of lags used; the weights $1 - h/(\ell+1)$ fall linearly

- **Phillips–Perron** (PP) Phillips and Perron (1988): the DF regression without lagged differences ($k = 0$)
  - the autocorrelation left in the residuals is handled by correcting τ with $\hat{\omega}^2$
  - same H_0 (unit root) and the same critical values as ADF

Prerequisites for Today (10/16): the KPSS test

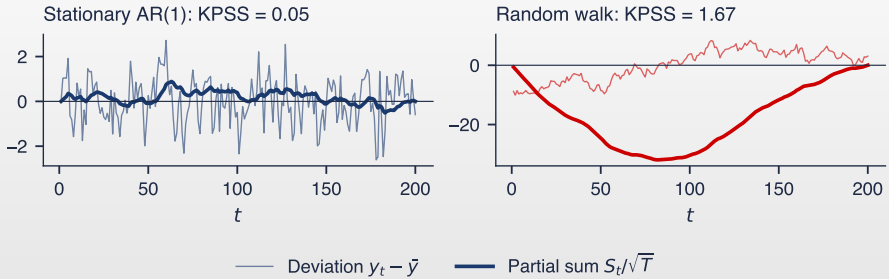
- **KPSS** Kwiatkowski et al. (1992): the null hypothesis is stationarity

$$\eta = \frac{1}{T^2 \hat{\omega}^2} \sum_{t=1}^{T} S_t^2, \qquad S_t = \sum_{s=1}^{t} \hat{e}_s$$

  - $\hat{e}_s$: the deviations from the mean (level) or from a fitted line (trend); S_t: their partial sums; $\hat{\omega}^2$: the long-run variance
  - H_0: stationary (around a level or a trend); H_1: unit root; reject for **large** values of η
  - 5% critical values: 0.463 (level), 0.146 (trend)

- **ADF and KPSS together**: opposite null hypotheses
  - ADF rejects, KPSS does not: $I(0)$; ADF does not, KPSS rejects: $I(1)$
  - neither rejects: inconclusive; both reject: conflict (often a break)

Prerequisites for Today (11/16): the KPSS partial sums



- Thin line: the deviations $y_t - \bar{y}$; thick line: their partial sums S_t, scaled by $\sqrt{T}$; $T = 200$
- Why large values reject stationarity
  - stationary AR(1), $\phi = 0.3$: positive and negative deviations cancel, S_t stays near 0; $\eta = 0.05 < 0.463$
  - random walk: long runs on one side of the mean, S_t makes a large swing; $\eta = 1.67 > 0.463$

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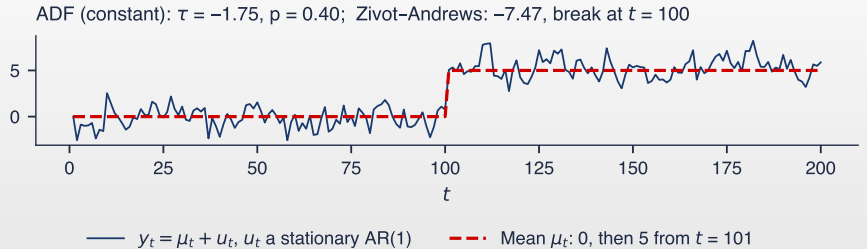
Prerequisites for Today (12/16): structural breaks and the Zivot–Andrews test

- **Structural break**: a shift in the mean or in the trend at a date T_B
  - a stationary series with one shift in its mean can look like a random walk to ADF
- **Zivot–Andrews** (ZA) Zivot and Andrews (1992): ADF with a break at an unknown date

$$\Delta y_t = c + \theta\, DU_t + bt + \gamma y_{t-\mathbf{1}} + \sum_{j=\mathbf{1}}^{k} \delta_j \Delta y_{t-j} + \varepsilon_t, \qquad DU_t = \begin{cases} 1, & t > T_B \\ 0, & t \le T_B \end{cases}$$

  - DU_t: a dummy variable that shifts the level by θ after T_B (a version also shifts the trend)
  - the regression is run for every candidate T_B; the statistic is the minimum τ over all dates
  - $H_{\mathbf{0}}$: unit root without a break; 5% critical values: -4.81 (break in the level), -5.07 (level and trend)

Prerequisites for Today (13/16): a break that looks like a unit root



- A simulated stationary AR(1), $\phi = 0.5$, whose mean jumps from 0 to 5 at $t = 101$; $T = 200$
- Two tests on the same series
  - ADF with a constant: $\tau = -1.75$, p $= 0.40$: the unit root is not rejected, although the series has none
  - Zivot–Andrews: $-7.47 < -4.81$: rejected; the estimated break, $t = 100$, is next to the true date

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Prerequisites for Today (14/16): ARIMA models and their forecasts

- **ARIMA**(p, d, q): an ARMA(p, q) for the series differenced d times

$$\phi(L)(1 - L)^d y_t = c + \theta(L)\varepsilon_t$$

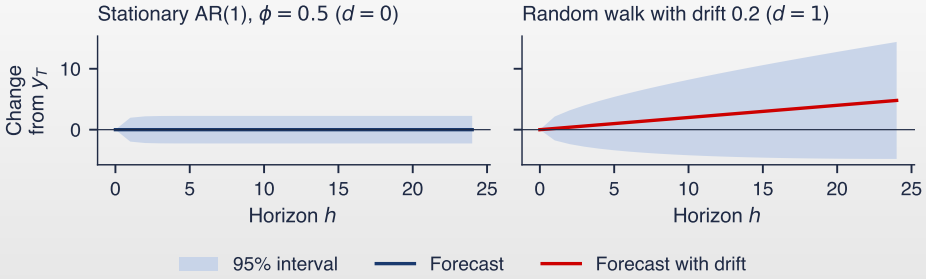
  - p: AR order; d: number of differences; q: MA order; $\phi(L)$, $\theta(L)$: the AR and MA polynomials (Chapter 2)
  - with $d = 1$, the constant c is the drift: the average change per period

- **Forecasts**: iterate the model for the levels, with future shocks set to 0

$$\text{Var}(y_{T+h} - \hat{y}_{T+h}) = \sigma^2(\psi_0^2 + \psi_1^2 + \cdots + \psi_{h-1}^2), \qquad \psi_0 = 1$$

  - ψ_j: the effect on y_{t+j} of a unit shock at t; σ^2: the variance of ε_t
  - ARIMA(1,1,0): $\psi_j = 1 + \phi + \cdots + \phi^j$; ARIMA(0,1,1): $\psi_j = 1 + \theta$ for $j \geq 1$; random walk: $\psi_j = 1$
  - interval $\hat{y}_{T+h} \pm z\sqrt{\text{Var}}$: $z = 1.96$ for 95%, $z = 1.28$ for 80% (quantiles of the standard Normal distribution)

Prerequisites for Today (15/16): forecast intervals with $d = 0$ and $d = 1$



- The 95% intervals of two models with $\sigma = 1$, as functions of the horizon h
- Stationary against integrated
  - AR(1), $\phi = 0.5$: $\sum_j \psi_j^2$ converges; the half-width stops at $1.96/\sqrt{1-\phi^2} = 2.26$
  - random walk: Var $= h\sigma^2$, the half-width $1.96\sqrt{h}$ grows without bound: 3.92 at $h = 4$, 7.84 at $h = 16$

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Prerequisites for Today (16/16): model choice and checks

- **AICc**: AIC (Chapter 2) with a small-sample correction; lower is better

$$\text{AICc} = -2 \ln L + 2k + \frac{2k(k+1)}{T-k-1}$$

  - $\ln L$: the maximised log-likelihood; k: the number of parameters; T: the number of observations
  - compare only models with the same d: differencing changes the data on which L is computed
- **Residual check**: Ljung–Box $Q^*(m)$ with $m - p - q$ degrees of freedom (Chapter 2)
- **Simple exponential smoothing** (SES, Chapter 0): a weighted average of the past

$$\hat{y}_{T+\mathbf{1}} = \alpha\, y_T + (1-\alpha)\, \hat{y}_T, \qquad 0 < \alpha < 1$$

  - $\hat{y}_T$: the forecast made for T one period earlier; α: the weight of the newest observation
  - the forecast is flat: $\hat{y}_{T+h} = \hat{y}_{T+\mathbf{1}}$ for every $h \geq 1$

A1: a Dickey–Fuller statistic [Solved]

Task

- A monthly interest rate, $T = 120$: OLS gives $\Delta y_t = 0.42 - 0.061\, y_{t-1}$, with $\mathrm{SE}(\hat{\gamma}) = 0.025$.
- 1. Compute τ and $\hat{\phi}$.
- 2. Compare τ with the 5% Dickey–Fuller value for a regression with a constant, -2.89, and decide.
- 3. Say what a standard one-sided t-test would have concluded.
- 4. Explain why a constant (and no trend) is the right choice for an interest rate.
- Report: two numbers, a decision and two sentences.

Solution

- 1. $\tau = -0.061/0.025 = -2.44$; $\hat{\phi} = 1 - 0.061 = 0.939$
- 2. $-2.44 > -2.89$: do not reject the unit root at 5% (p = 0.13); also not at 10% (-2.58)
- 3. $-2.44 < -1.645$: the t-test would wrongly reject; under H_0 the statistic is not Student
- 4. An interest rate has no long-run trend; the plausible alternative is "stationary around a mean", which the constant represents.

Back Exercise list

A2: choosing the deterministic terms [Proposed]

Solution discussed in class

Task

- The log of a price index ($T = 200$) rises steadily. ADF: $\tau = 1.85$ (no constant), -1.21 (constant), -3.62 (constant and trend); KPSS with trend: 0.09. Model: A1.
- 1. Say which specification fits the data, and why.
- 2. Decide at 5% in each specification, with the critical values -1.94 (no constant), -2.88 (constant) and -3.43 (constant and trend).
- 3. Combine the chosen ADF result with KPSS (5% value 0.146).
- 4. Say how you would model the series.
- Report: three decisions, a verdict and one sentence.

Back Exercise list

A3: ARIMA(1,1,0) forecasts [Solved]

Task

- $\Delta y_t = 0.6\,\Delta y_{t-\mathbf{1}} + \varepsilon_t$, $\sigma = 2$; the last two values are $y_{T-\mathbf{1}} = 103$ and $y_T = 108$.
- 1. Compute $\hat{y}_{T+\mathbf{1}}$, $\hat{y}_{T+\mathbf{2}}$ and $\hat{y}_{T+\mathbf{3}}$.
- 2. Compute $\psi_{\mathbf{1}}$ and $\psi_{\mathbf{2}}$, and the forecast error variances for $h = 1, 2, 3$.
- 3. Give the 95% intervals.
- Report: three forecasts and three intervals.

Solution

- 1. $\Delta y_T = 5$; $\widehat{\Delta y}_{T+\mathbf{1}} = 3$, $\hat{y}_{T+\mathbf{1}} = 111.00$; $\widehat{\Delta y}_{T+\mathbf{2}} = 1.8$, $\hat{y}_{T+\mathbf{2}} = 112.80$; $\widehat{\Delta y}_{T+\mathbf{3}} = 1.08$, $\hat{y}_{T+\mathbf{3}} = 113.88$
- 2. $\psi_{\mathbf{1}} = 1.6$, $\psi_{\mathbf{2}} = 1.96$; variances 4, $4(1 + 2.56) = 14.24$, $4(1 + 2.56 + 3.8416) = 29.61$
- 3. $[107.08, 114.92]$; $[105.40, 120.20]$; $[103.22, 124.54]$: half-widths 3.92, 7.40, 10.66

Back Exercise list

A4: ARIMA(0,1,1) and exponential smoothing [Proposed]

Solution discussed in class

Task

- $\Delta y_t = \varepsilon_t - 0.6\,\varepsilon_{t-\mathbf{1}}$, $\sigma = 1$; $y_T = 50$ and the last residual is $\hat{\varepsilon}_T = 1.5$. Model: A3.
- 1. Compute $\hat{y}_{T+\mathbf{1}}$ and $\hat{y}_{T+h}$ for $h \geq 2$.
- 2. Compute the forecast error variances for $h = 1$, $h = 2$ and $h = 5$, and the 95% intervals.
- 3. Show that the forecast equals simple exponential smoothing (Chapter 0) and give α.
- Report: two forecasts, three intervals and α.

A5: the order of integration of an equation [Solved]

Task

- $y_t = 1.5\, y_{t-\mathbf{1}} - 0.5\, y_{t-\mathbf{2}} + \varepsilon_t + 0.4\, \varepsilon_{t-\mathbf{1}}$.
- 1. Write the AR polynomial $\phi(z)$ and find its roots.
- 2. Say whether y_t is stationary, and give d.
- 3. Write the model as ARIMA(p, d, q) for Δy_t.
- Report: two roots, d and the ARIMA equation.

Solution

- 1. $\phi(z) = 1 - 1.5z + 0.5z^{\mathbf{2}} = (1 - z)(1 - 0.5z)$: roots $z = 1$ and $z = 2$
- 2. One root on the unit circle: not stationary; the other root is outside: $d = 1$
- 3. $(1 - 0.5L)\Delta y_t = (1 + 0.4L)\varepsilon_t$: ARIMA(1,1,1), $\Delta y_t = 0.5\, \Delta y_{t-\mathbf{1}} + \varepsilon_t + 0.4\, \varepsilon_{t-\mathbf{1}}$

Back Exercise list

A6: integrated or over-differenced? [Proposed]

Solution discussed in class

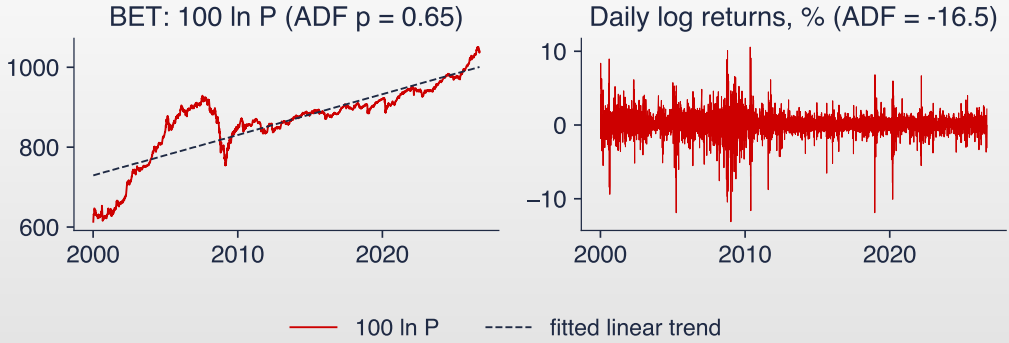
Task

- Two models: (a) $y_t = 1.8\,y_{t-\mathbf{1}} - 0.8\,y_{t-\mathbf{2}} + \varepsilon_t$; (b) $\Delta y_t = \varepsilon_t - \varepsilon_{t-\mathbf{1}}$. Model: A5.
- 1. For (a), factor $\phi(z)$, give d and write the ARIMA model.
- 2. For (b), find the MA root and say what y_t really is.
- 3. Compute $\rho(1)$ of Δy_t in (b) and say how you would detect this case in data.
- Report: two orders of integration and two sentences.

B1: is the BET log price a random walk? [Solved]

- **Question**: does the BET log price have a unit root, and are its daily returns stationary?
- **Inputs**: BET closes since 2000; $p_t = 100 \ln P_t$ and $r_t = \Delta p_t$
- **Tasks**
  1. Plot p_t with a fitted linear trend, and plot r_t.
  2. Run ADF (lags by AIC), PP and KPSS on p_t with constant and trend.
  3. Run the same tests on r_t with a constant.
  4. Interpretation: if the BET is far below its fitted trend line, should you expect it to return to the line?
- **Report**: a table of six statistics, two verdicts and two sentences
- Notebook: section B1

B1: solution (1/2) [Solved]



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B1: solution (2/2) [Solved]

	ADF (p, k)	PP	KPSS	**verdict**
p_t (c, t)	-1.90 (0.653, 19)	-2.35	1.143	$I(1)$
r_t (c)	-16.50 (<0.001, 18)	-74.55	0.360	$I(0)$

- p_t: ADF and PP far above -3.41, KPSS > 0.146: unit root; r_t: both reject strongly, KPSS < 0.463: stationary
- Interpretation: no; with a unit root the trend line is not an attractor; at the end of the sample the index is 39 log points from the line, and nothing pulls it back

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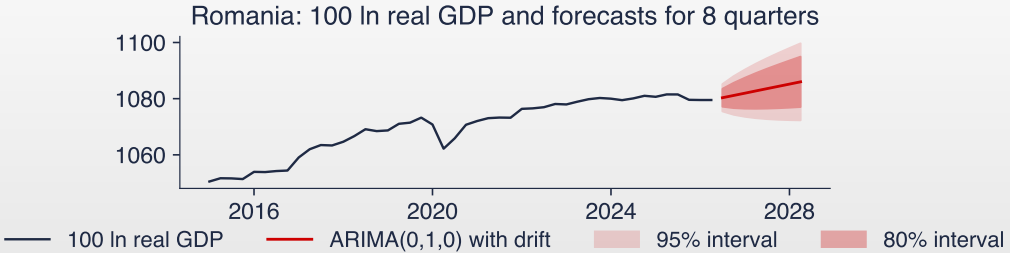
B2: the S&P 500 and EUR/RON [Proposed]

- **Question**: do the S&P 500 and the EUR/RON rate give the same verdicts as the BET?
- **Inputs**: S&P 500 since 2000; EUR/RON (BNR reference rate) since July 2005; both as $100 \ln$; model: B1
- **Tasks**
  1. Repeat steps 2 and 3 of B1 for both series.
  2. Compare the ADF and PP statistics of the returns, and explain why PP is much more negative.
  3. Interpretation: the EUR/RON rate is managed by the central bank; does the test result say that it is a random walk?
- **Report**: a table (two series, two rows each) and two sentences
- Notebook: section B2

B3: an ARIMA model for Romanian GDP [Solved]

- **Question**: which ARIMA model describes Romanian real GDP, and how uncertain is a two-year forecast?
- **Inputs**: real GDP, chain-linked volumes (2010), seasonally adjusted, Eurostat, from 2000Q1; $y_t = 100 \ln Y_t$
- **Tasks**
  1. Choose d with ADF and KPSS on y_t (constant and trend) and on Δy_t (constant).
  2. Fit ARIMA$(p, 1, q)$ with drift for $p, q \leq 2$ and choose the model with the lowest AICc.
  3. Check the residuals with Ljung–Box $Q^*(8)$.
  4. Forecast 8 quarters with 80% and 95% intervals.
  5. Interpretation: why is the 95% interval after 8 quarters about twice as wide as after 2 quarters?
- **Report**: the tests, the AICc of six models, the chosen model, $Q^*(8)$, the chart and one sentence
- Notebook: section B3

B3: solution (1/2) [Solved]



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B3: solution (2/2) [Solved]

- y_t: ADF -2.31 (p $= 0.428$), KPSS $0.198 > 0.146$: $I(1)$; Δy_t: ADF -10.77, KPSS 0.263: $I(0)$; so $d = 1$
- AICc: (0,1,0) 491.9; (1,1,0) 493.8; (0,1,1) 493.7; (1,1,1) 494.9; (2,1,0) 493.7; (0,1,2) 493.6: ARIMA(0,1,0) with drift $\hat{c} = 0.81$ (SE 0.30), $\hat{\sigma} = 2.47$
- $Q^*(8) = 7.3$ (p $= 0.51$): white-noise residuals; forecast for 2028Q2: 1086.0, i.e. $+6.5\%$ in two years, 95% interval $[1072.3, 1099.7]$
- Interpretation: for a random walk the error variance is $h\sigma^2$, so the width grows like $\sqrt{h}$: $\sqrt{8/2} = 2$ (half-widths 6.8 and 13.7, ratio 2.00)

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B4: Romanian inflation, $d = 0$ or $d = 1$? [Proposed]

- **Question**: is Romanian 12-month inflation stationary, and how much does the answer change the forecast?
- **Inputs**: π_t, the 12-month inflation of the data slide, HICP, January 2005 to August 2026; model: B3
- **Tasks**
  1. Run ADF, PP and KPSS on π_t and on $\Delta\pi_t$, with a constant.
  2. Find the best ARIMA$(p, 0, q)$ with a mean and the best ARIMA$(p, 1, q)$ by AICc, with $p \leq 3$, $q \leq 2$.
  3. Forecast 12 months with both models, with 95% intervals, and run Ljung–Box $Q^*(24)$ on their residuals.
  4. Interpretation: why do the two models give similar point forecasts but different intervals?
- **Report**: a table of tests, two models, two forecasts with intervals, two Ljung–Box tests and two sentences
- Notebook: section B4

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C1: a break or a unit root? [Proposed]

- **Question**: do Romanian inflation and GDP have a unit root, or are they stationary around a mean or a trend that shifted once?
- **Inputs**: π_t of B4 and y_t of B3; models: B1, B3
- **Tasks**
  1. Run the Zivot–Andrews test with a break in the level, then in the level and the trend, on both series.
  2. Compare each statistic with its 5% critical value and report the break dates.
  3. Match the dates with events (inflation targeting since 2005, the 2008–2009 crisis, the 2022 energy shock).
  4. Interpretation: why should a break date found by a test be checked against history before it is believed?
- **Report**: a table of four statistics with dates, the chart and a plan for a project
- Notebook: section C1

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C2: audit an AI answer [Proposed]

- A student asked an AI assistant to analyse Romanian GDP and prices. The answer:
- (a) The ADF p-value for log GDP is 0.43, so GDP is proved to have a unit root.
- (b) The KPSS statistic for log GDP is 0.198, above 0.146, so KPSS rejects the unit root.
- (c) Regressing log HICP on the log S&P 500 gives t = 34.6 and R2 = 0.82: US stocks drive Romanian prices.
- (d) GDP growth is stationary, so differencing it once more will make the ARIMA model even better.
- (e) A random-walk forecast has the same 95% interval width at every horizon.
- (f) The 5% Dickey-Fuller critical value with a constant is about -2.87, more negative than -1.645, because the Dickey-Fuller distribution is shifted to the left.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - 2. Report: a list of six verdicts with one line of justification each.

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Key takeaways

- Compare τ with Dickey–Fuller critical values, never with Normal ones; choose the deterministic terms from the plot
- ADF and KPSS have opposite null hypotheses: read them together
- Prices, exchange rates and GDP in logs are $I(1)$; returns and growth rates are $I(0)$
- With $d = 1$, forecast intervals grow with the horizon; the choice of d matters most for long horizons
- An AI answer is a draft: check the direction of each test and the critical values used

After the seminar

- Lecture 3 develops each topic of today: trends, spurious regression, the Dickey–Fuller distributions, PP, KPSS, breaks, ARIMA forecasting and automatic selection
- C1 can grow into a team project: breaks and persistence of inflation in Romania, Hungary, Poland and Czechia
- Reading: Huang and Petukhina (2022), Ch. 4–5; Hyndman and Athanasopoulos (2021), Ch. 9; Hamilton (1994), Ch. 15–17

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